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Homogeneous Catalysis

Chromium-Catalyzed Selective Dimerization/Hydroboration of Allenes to Access Boryl-Functionalized Skipped (*E*,*Z*)-Dienes

*Yinsong Zhao and Shaozhong Ge**

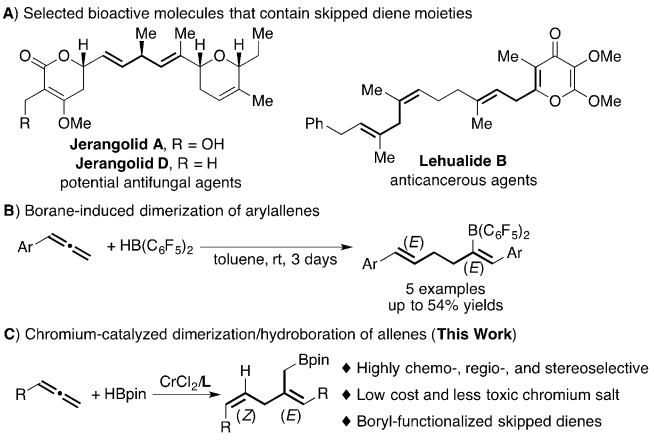
Abstract: *A chromium-catalyzed dimerization/hydroboration of allenes is developed to access synthetically versatile boryl-functionalized skipped dienes with a catalyst generated in situ from $CrCl_2$ and a pyridine-2,6-diimine ligand ^{mes}PDI. A variety of allenes reacted with pinacolborane (HBpin) to afford the corresponding boryl-functionalized (E,Z)-1,4-dienes in high yields and with excellent selectivity. Electron paramagnetic resonance (EPR) spectroscopic studies suggest that this chromium-catalyzed reaction probably proceeds through a chromium(I) hydride intermediate.*

Catalysts based on first-row transition metals, such as iron, cobalt, nickel, and copper, have been extensively studied in reaction discovery and development over past decades partially because of their low price, high abundance, and low toxicity.[1] Chromium, a first-row early transition metal in Group 6, is an abundant element in the Earth's crust. Although Cr^{VI} compounds are highly toxic, Cr^{II} and Cr^{III} compounds, such as $CrCl_2$ and $CrCl_3$, have low toxicity.[2] Chromium compounds have shown unique catalytic reactivities in Nozaki–Hiyama–Kishi reactions,[3] various cross-coupling reactions,[4] and the polymerization and oligomerization of olefins.[5] However, chromium-catalyzed hydroboration reactions of unsaturated hydrocarbons that can afford synthetically versatile organoboron compounds[6] have remained unknown.

Skipped dienes are valuable structural motifs in numerous bioactive natural products and synthetic compounds (Scheme 1 A).[7] Several synthetic approaches, such as hydroallylation of alkynes,[8] hydroalkenylation of 1,3-dienes,[9] and coupling of vinyl metallic reagents with various allylic electrophiles,[10] have been developed to prepare skipped dienes. However, systematic approaches to access boryl-functionalized skipped dienes from readily accessible starting materials are still lacking. Considering broad applications of organoboron compounds in chemical synthesis,[6] we are interested in developing selective approaches to synthesize boryl-functionalized skipped dienes because they can be further converted to other unsaturated hydrocarbons via selective C–B bond transformations.

Allenes are polyenes containing two cumulative carbon-carbon double bonds, and they can undergo dimerization or trimerization in the presence of transition-metal catalysts.[11] Therefore, we envisioned that a reaction between an allene and HBpin with a stoichiometry of 2:1 would afford a boryl-functionalized diene, provided that a catalyst could be identified to catalyze sequential dimerization and hydroboration of allenes. Catalytic hydroboration of allenes has been well studied with various metal catalysts to prepare allylic or alkenyl boronates.[12] Nevertheless, none of these catalysts have been shown to promote dimerization of allenes during hydroboration reactions. There is one example of a borane-mediated dimerization of arylallenes reported by Erker (Scheme 1 B),[13] and this stoichiometric reaction employs a highly reactive and electrophilic hydroborane $(C_6F_5)_2BH$ and affords a boryl-functionalized (*E*,*E*)-1,5-diene product.

During our recent studies on cobalt-catalyzed hydrofunctionalization of allenes,[12d,14] we became interested in exploring other base metal catalysts for the reactions of allenes with pinacolborane (HBpin), with the aim of discovering new selectivity. Herein, we report the first selective chromium-catalyzed dimerization/hydroboration of allenes with HBpin to prepare boryl-functionalized skipped (*E*,*Z*)-dienes (Scheme 1 C). We also show that these boryl-containing (*E*,*Z*)-1,4-diene products can be converted to various stereo-defined alkenyl compounds by standard functional group interconversions. Furthermore, preliminary mechanistic studies including deuterium-labeling and electron paramagnetic resonance (EPR) spectroscopic experiments suggest that this chromium-catalyzed dimerization/hydrobora-



Scheme 1. Boryl-functionalized skipped dienes generated by coupling of two allenes with boron reagents.

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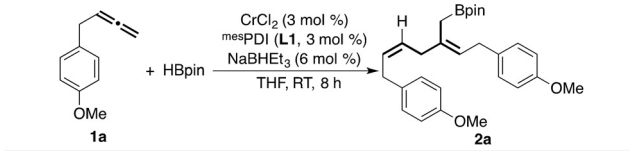
tion reaction occurs through a chromium(I) hydride intermediate.

We initiated our studies on dimerization/hydroboration of allenes by identifying a selective catalyst and reaction conditions for hydroboration of allene **1a** with HBpin to yield the 1,4-diene product **2a** (Table 1). After systematic evaluations of various first row metal catalysts (entry 12 in Table 1) and other reaction parameters, we found that the reaction conducted with the catalyst generated from $CrCl_2$ and a pyridine-2,6-diimine ligand ^{mes}PDI (**L1**) afforded the desired product **2a** in 78% isolated yield (entry 1 in Table 1). The reaction did not occur in the absence of $CrCl_2$ (entry 2 in Table 1), ligand **L1** (entry 3 in Table 1), or the activator $NaBHEt_3$ (entry 4 in Table 1). We also tested other PDI ligands **L2**-**L7** for this transformation, but these reactions produced **2a** either in decreased yields (entries 5 and 6 in Table 1) or only in trace amounts (entries 7 and 8 in Table 1). The catalyst generated from $CrCl_3$ and **L1** was also active for this transformation (entry 9 in Table 1). We also tested other activating reagents for this reaction. For example, the reaction run with EtMgBr occurred to complete conversion of **1a**, but the yield of **2a** for this reaction is slightly decreased when compared with the reaction conducted with $NaBHEt_3$ (64%; entry 10 in Table 1). The reactions run with NaH, magnesium, or zinc as activators did not take place (entry 11 in Table 1).

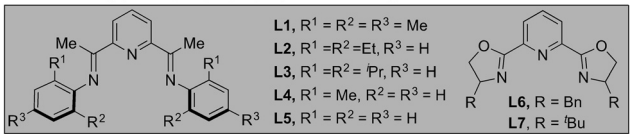
With the optimized reaction condition in hand (entry 1 in Table 1), the scope of allenes was investigated (Table 2). Generally, a variety of alkyl- and aryl-substituted allenes reacted with HBpin in the presence of 3 mol% $CrCl_2$/^{mes}PDI and 6 mol% $NaBHEt_3$, affording the corresponding boryl-containing skipped (*E*,*Z*)-dienes (**2a**-**2z**) in modest to high isolated yields (52%-87%). This Cr-catalyzed transformation shows good functional group tolerance, and various reactive groups, such as ether (**2g**-**2k**), fluoro (**2d**), chloro (**2g** and **2z**), bromo (**2e**, **2h**, and **2k**), acetal (**2j**), tosylate (**2q**), silyl ether (**2r** and **2s**), terminal alkene (**2i** and **2t**), and thioether (**2y**) moieties are compatible with the reaction conditions. Furthermore, silyl-substituted and heteroaryl-containing allenes also reacted to afford the corresponding dienes **2o** and **2v** in good isolated yields, respectively. However, allenes containing ketone or aldehyde groups do not undergo this Cr-catalyzed transformation. Furthermore, we also conducted a gram-scale reaction with allene **1p** (6.00 mmol), and this reaction afforded the diene product **2p** (1.03 g) in 72% isolated yield.

We then explored Cr-catalyzed dimerization/hydroboration with two different allenes. Initial attempts on the

Table 1: Optimization of reaction conditions.[a]

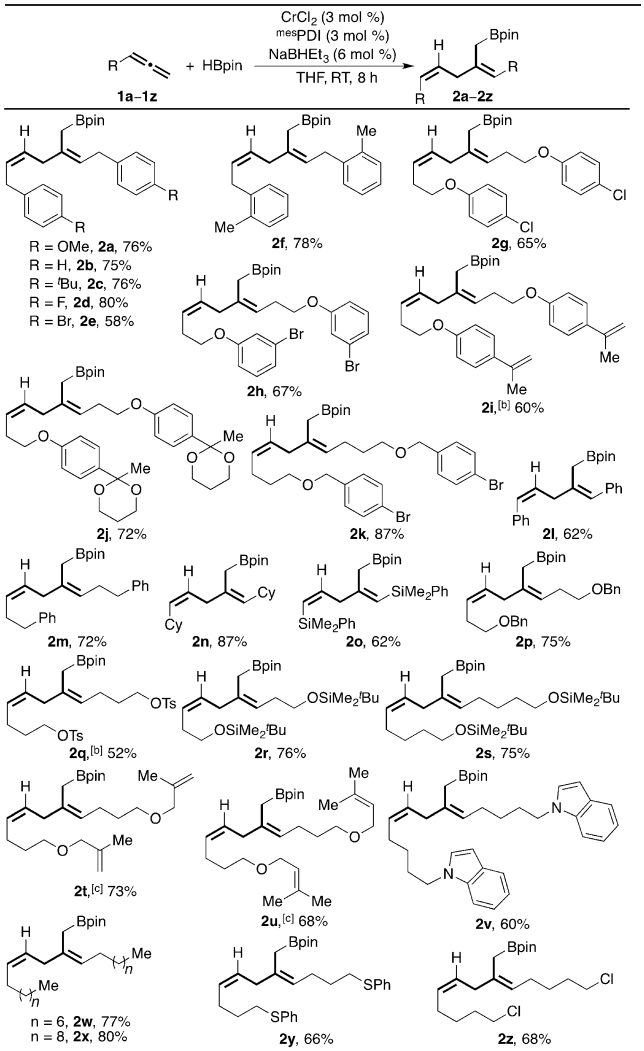


entry	Variation from the standard conditions	conv. of **1a** (%)	yield of **2a** (%)
1	none	>99	78
2	without $CrCl_2$	--	--
3	without **L1**	--	--
4	without $NaBHEt_3$	--	--
5	**L2** instead of **L1**	>99	65[e]
6[b]	**L3** instead of **L1**	>99	23
7[c]	**L4** or **L5** instead of **L1**	52	<5
8[c]	**L6** or **L7** instead of **L1**	50	<5
9[d]	$CrCl_3$ instead of $CrCl_2$	>99	73
10	EtMgBr instead of $NaBHEt_3$	>99	64[e]
11	NaH, Mg, and Zn powder intead of $NaBHEt_3$	<5	<5
12[f]	MCl_2 (M = Mn, Fe, Co, Ni, Cu) intead of $CrCl_2$	30–50	<5



[a] Conditions: **1a** (0.200 mmol), HBpin (0.120 mmol), $CrCl_2$ (3.0 μmol), ligand (3.0 μmol), activator (6.0 μmol), THF (0.2 mL), RT, 8 h, isolated yields. [b] Product from hydrodimerization of **1a** was the major product. [c] Hydroboration of **1a** occurred predominantly. [d] $NaBHEt_3$ (9.0 μmol) was used. [e] This reaction also afforded some products of hydrodimerization of **1a**. [f] These reactions afforded a mixture of products from hydrogenation, hydroboration, and hydrodimerization reactions of **1a**.

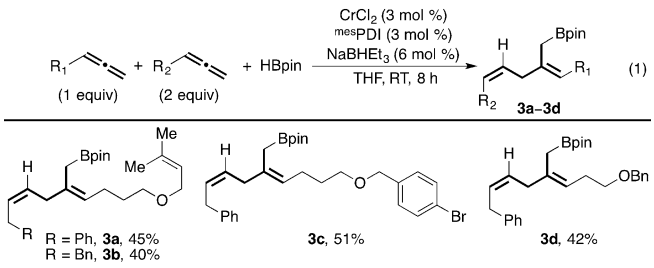
Table 2: Scope of allenes for Cr-catalyzed dimerization/hydroboration.[a]



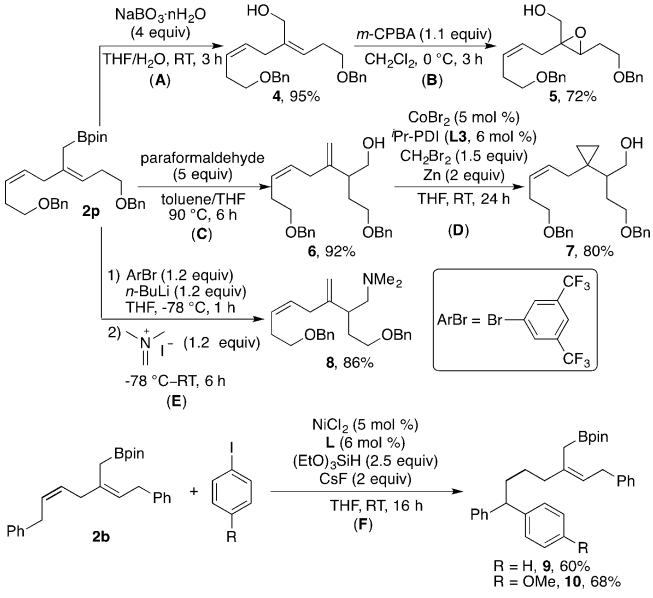
[a] Conditions: allene (1.00 mmol), HBpin (0.600 mmol), $CrCl_2$ (15.0 μmol), ^{mes}PDI (15.0 μmol), $NaBHEt_3$ (30.0 μmol), THF (1 mL), RT, 8 h, and isolated yields. [b] 50°C. [c] 40°C.

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reactions employing two allenes with equal molar ratio showed that cross-dimerization/hydroboration products did not form with noticeable selectivity. However, increasing the molar ratio of two allenes to 1:2 led to an increased selectivity that allows effective purification of cross-dimerization/hydroboration products. The selected examples of cross-dimerization/hydroboration reactions are given in Equation (1), and these reactions afforded the desired diene products with good selectivity (with the ratios of products from cross- and homodimerization/hydroboration varying from 6:1 to 15:1), albeit in modest isolated yields.



To demonstrate the synthetic utility of this chromium-catalyzed dimerization/hydroboration procotol, we conducted various synthetic transformations with the boryl-functionalized diene products **2b** and **2p**, and these reactions are summarized in Scheme 2. Compound **2p** could be readily oxidized by $NaBO_3$, affording allylic alcohol **4** in 95% isolated yield (Scheme 2A). Epoxidation of compound **4** with *m*-CPBA occurred selectively on the trisubstituted alkene unit to produce epoxide **5** in 72% yield (Scheme 2B). Allylic boronate **2p** could undergo hydroxymethylation with paraformaldehyde to afford homoallylic alcohol **6**, which could be converted to compound **7** selectively in high yield via cobalt-catalyzed cyclopropanation of alkenes (Scheme 2C,D).[15] Compound **2p** also underwent aminomethylation to produce homoallylic amine **8** in 86% yield (Scheme 2E).[16] Furthermore, we also showed that diene product **2b** could undergo NiH-catalyzed remote hydroarylation with aryl iodides in the presence of $(EtO)_3SiH$ to give allylic boronates **9** and **10** in good isolated yields (Scheme 2F).[17]



Scheme 2. Transformations of boryl-functionalized skipped dienes **2b** and **2p**.

To provide insights into the nature of the active chromium species for this reaction, we conducted electron paramagnetic resonance (EPR) spectroscopic studies on the chromium catalyst (Figure 1). No EPR signals were observed for the mixture of $CrCl_2$ and ^{mes}PDI ligand with a molar ratio of 1:1. However, upon addition of 2 equiv of $NaBHEt_3$ to the mixture of $CrCl_2$ and ^{mes}PDI, a singlet resonance appeared with a *g* value of 2.001 (Figure 1A), which is consistent with the value observed for Cr^I species.[5f] We then monitored by EPR spectroscopy the reaction of allene **1a** with HBpin catalyzed by $CrCl_2$/^{mes}PDI in the presence of $NaBHEt_3$ (entry 1 in Table 1), and observed a singlet resonance with a *g* value of 1.998 (Figure 1B). A similar singlet resonance with a *g* value of 1.999 was observed for the corresponding dimerization/hydroboration reaction catalyzed by $CrCl_3$/^{mes}PDI (entry 9 in Table 1). The results of these EPR spectroscopic studies suggest that Cr^I intermediates are likely the active species in this Cr-catalyzed transformation.

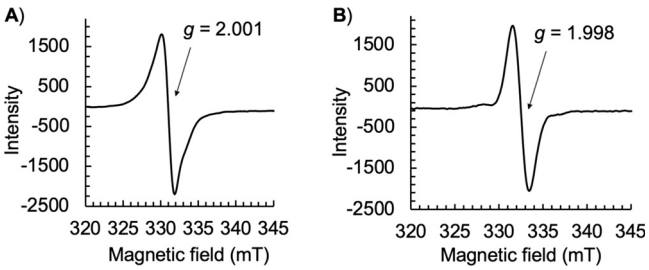
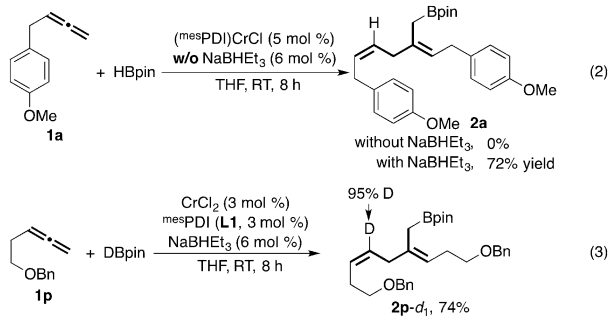


Figure 1. EPR Spectra of chromium species.

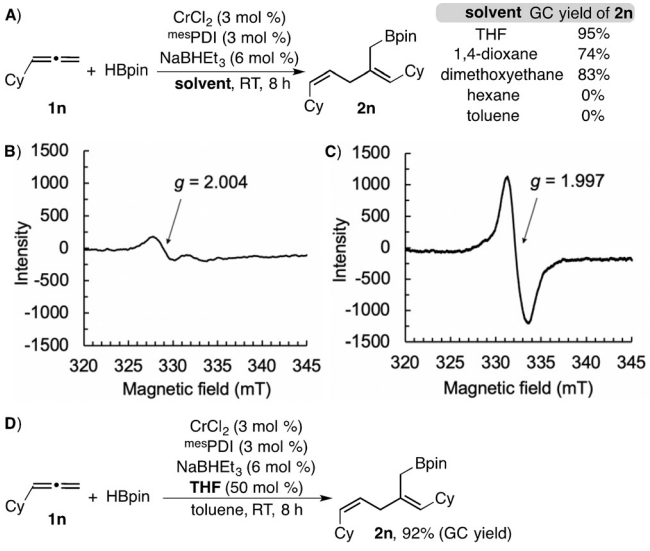
We subsequently conducted some control experiments to get more information about the active chromium intermediates. Previous studies showed that PDI-ligated $CrCl_2$ could be reduced to PDI-ligated CrCl compound by NaH.[5e] However, the reaction of allene **1a** with HBpin catalyzed by $CrCl_2$/^{mes}PDI in the presence of NaH did not proceed (entry 11 in Table 1), probably because of sluggish activation of the catalyst precursor by NaH. We then carried out this reaction with preformed (^{mes}PDI)CrCl, and this reaction still did not occur [Eq. (2)]. However, the combination of (^{mes}PDI)CrCl and 1.2 equivalents of $NaBHEt_3$ was active for dimerization/hydroboration of allene **1a** [Eq. (2)]. We then conducted EPR spectroscopic analysis on the reaction catalyzed by (^{mes}PDI)CrCl and $NaBHEt_3$, and observed a singlet resonance with a *g* value of 1.998 (Supporting Information, Figure S5), which is the same with the resonance shown in Figure 1B. Furthermore, we also performed a deuterium-labeling experiment for the reaction of allene **1p** with DBpin under standard

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conditions [Eq. (3)]. This reaction afforded compound **2p**-d_1 in 74% yield, and the deuterium atom was selectively incorporated onto the (*Z*)-alkenyl carbon. These results suggest that a Cr^{I}–H species is likely the active chromium intermediate of this dimerization/hydroboration reaction.



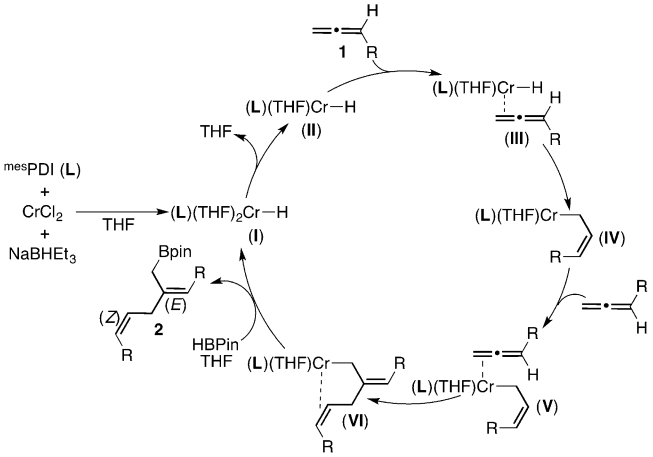
We also found that this chromium-catalyzed dimerization/hydroboration of allenes showed profound solvent effects (Scheme 3A). For example, the reaction between cyclohexyl-substituted allene **1n** and HBpin proceeded to full conversion of **1n** and yielded the desired boryl-containing skipped diene **2n** in high yields (74–95%, Scheme 3A) when these reactions were conducted in solvents with high coordinating ability, such as THF, 1,4-dioxane, and dimethoxyethane. However, the corresponding reactions conducted in non-coordinating solvents, such as toluene and hexane, did not take place (Scheme 3A). We subsequently monitored by EPR spectroscopy the reaction of allene **1n** with HBpin catalyzed by 3 mol% of $CrCl_2$/^{mes}PDI in toluene and the EPR spectrum showed a very weak signal with a *g*-value of 2.004 (Scheme 3B). However, the addition of 0.5 equivalent of THF, relative to allene **1n**, to the reaction mixture promoted the formation of a significant amount of active chromium species (Scheme 3C). Accordingly, the reaction of allene **1n** with HBpin catalyzed by 3 mol% of $CrCl_2$/^{mes}PDI in toluene also proceeded to full conversion of **1n** and afforded the desired product in 92% GC yield when 50 mol% THF was added (Scheme 3D). To account for these observed solvent effects, we rationalized that coordinating solvents might help to solvate $CrCl_2$ and act as a ligand to stabilize the sterically and electronically unsaturated Cr^{I} species.



Scheme 3. Solvent effects of the reaction between **1n** and HBpin.

The kinetic behavior of this chromium-catalyzed dimerization/hydroboration of allenes was assessed by conducting kinetic studies on the reaction of cyclohexyl-substituted allene **1n** with HBpin catalyzed by $CrCl_2$ and ^{mes}PDI in THF at room temperature (Supporting Information, Figure S8). The plots of the initial rates of the reaction vs. the concentration of the catalyst, allene **1n**, and HBpin are shown in the Supporting Information, Figures S9–S11. The results of these kinetic studies showed that this chromium-catalyzed dimerization/hydroboration of allene is first order in chromium catalyst and zero order in allene and HBpin.

Based on the results of EPR spectroscopic analysis, control experiments, solvent effects, and kinetic studies, we proposed a catalytic cycle depicted in Scheme 4 for this chromium-catalyzed dimerization/hydroboration of allenes. The activation of $CrCl_2$ with $NaBHEt_3$ in the presence of ^{mes}PDI in THF generates a Cr^{I} hydride (**L**)$(THF)_2$Cr–H (**I**). Dissociation of a THF molecule from the chromium intermediate **I** forms a sterically and electronically unsaturated Cr^{I} hydride species (**L**)(THF)Cr–H (**II**). Allene **1** then coordinates with (**L**)(THF)Cr–H followed by migratory insertion to generate an allylchromium intermediate (**IV**).[18] Subsequent reactions of allene **1** with allylchromium species (**IV**) via coordination and insertion form a C–C bond between two allene molecules and generate a new allylchromium species (**VI**).[19] The intramolecular coordination of the *cis*-C=C bond in this chromium intermediate might inhibit its further migratory insertion reactions with allenes. In the last step, intermediate **VI** reacts with HBpin in THF to afford product **2** and regenerate Cr^{I} hydride species (**L**)$(THF)_2$Cr–H. In the chromium intermediates (**III**) and (**V**), the allene coordinates with chromium catalysts in a way that the substituent R points away from the catalysts to minimize the steric repulsion, which explains the stereochemistry of the two double bonds in the skipped (*E*,*Z*)-diene product **2**. The observed first order



Scheme 4. Proposed catalytic cycle.

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rate dependence on the concentration of the chromium catalyst and zero order rate-dependence on the concentrations of allene and HBpin indicate that the dissociation of THF from (**L**)(THF)$_2$Cr–H (**I**) to form (**L**)(THF)Cr–H (**II**) is the turnover-limiting step for this dimerization/hydroboration process.

In summary, we have developed an effective protocol to prepare boryl-functionalized skipped (*E*,*Z*)-1,4-dienes via selective chromium-catalyzed dimerization/hydroboration of allenes. Various terminal allenes reacted with HBpin to produce the corresponding boryl-containing skipped dienes in high yields with excellent regio-, chemo-, and stereoselectivity. This reaction provides a versatile access to series of stereodefined alkenyl and skipped dienyl compounds. Preliminary mechanistic studies suggest that this chromium-catalyzed dimerization/hydroboration occurs through a Cr^{I}–H intermediate. Studies to determine the detailed mechanism of this transformation and to develop chromium-catalyzed hydrofunctionalization of other types of unsaturated hydrocarbons will be the subjects of future work.

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Conflict of interest

The authors declare no conflict of interest.

Keywords: allenes · chromium · dimerization · homogeneous catalysis · hydroboration

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